NUTM

NIGERIAN UNIVERSITY OF TECHNOLOGY & MANAGEMENT

CSC 309 · 2 UNITS · CORE

Artificial Intelligence

By

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COURSE OUTLINE



Welcome to CSC 309!

You will learn how machines can search, reason, understand language, see and act — and you will build working AI programs yourself.

What this course is about

- **What AI is**: its history, goals, techniques, types, branches and applications.
- **Intelligent agents**: performance, rationality, environments and architectures.
- **Problem solving by search**: search strategies, heuristics, games and combinatorial explosion.
- **Knowledge representation**: logic, frames and reasoning under uncertainty.
- **Natural language processing**: understanding language.
- **Expert systems**, **programming languages for AI**, and **computer image recognition**.
- Practicals: Turing test, face recognition, NLP apps, agent simulations, AI animation, simple robots.

Learning outcomes (1–4)

1

Explain AI fundamentals

Concepts, goals, types, techniques, branches, applications, technology and tools.

2

Discuss intelligent agents

Performance, examples, faculties, environments, architectures; characterise a problem.

3

Describe the Turing test and the Chinese Room

And contrast optimal reasoning with human-like reasoning.

4

Describe the role of heuristics

And the trade-offs between completeness, optimality, time and space.

Learning outcomes (5–8)

1

Analyse types of search and their applications

And explain combinatorial explosion and its consequences.

2

Demonstrate knowledge representation

Semantic networks and frames, and where to use them.

3

Practise natural language processing

English → predicate logic → clause form → resolution to answer a query.

4

Analyse AI programming languages and expert systems

And apply AI in real application domains.

Why AI matters — for you and for Nigeria



Health

Faster diagnosis, e.g. reading scans and test results.



Farming

Crop-disease detection and weather advice for farmers.



Finance

Fraud detection and credit scoring in banks and fintechs.



Languages

Tools for Hausa, Igbo, Yoruba and Pidgin speakers.



Careers

AI and data skills are among the most in-demand in the world.



Responsibility

AI must be fair, safe and respect privacy.

The course map: 5 modules



Module 1

Foundations of AI

- History, goals, types, branches
- Applications, pros and cons
- Turing test, Chinese Room

YOU WILL BUILD

ML Models



Module 2

Intelligent agents

- Performance & rationality
- Environments & architectures
- Multi-agent simulations

YOU WILL BUILD

Ant colony and flocking simulations



Module 3

Search & heuristics

- Uninformed & informed search
- Heuristics, games
- Combinatorial explosion

YOU WILL BUILD

A route finder; game AI



Module 4

Knowledge & reasoning

- Logic, clause form, resolution
- Semantic nets & frames
- Uncertainty & Bayes

YOU WILL BUILD

A resolution prover; frames



Module 5

NLP, experts, vision, robots

- NLP practicals
- Expert systems; Lisp & Prolog
- Vision, robots, lawn robot

YOU WILL BUILD

Spam filter, face recognizer, Naira recognizer



Alan Turing (1912–1954)

Module 1 – Foundations of AI

- What AI is: the four approaches; AI, machine learning and deep learning.
- **History**: from Turing and Dartmouth (1956) to ChatGPT — and Nigeria's AI journey.
- Goals, the **AI technique**, types and **branches** of AI; applications; advantages and disadvantages.
- The **Turing test**, the **Chinese Room**, and **optimal vs human-like** reasoning.

Module 1 – What you will do



Turing test practical

Act out your own Turing test in class.



Build ML Models



Monty Hall

Simulate it: optimal vs human intuition.



Debate

Can machines think? The Chinese Room.



An ant trail: simple agents, clever teamwork

Module 2 – Intelligent agents and environments

- Agents, percepts and actions; the agent function and program.
- **Performance**, **rationality**, and the **faculties** of an agent.
- **Environments**: seven properties; determining the characteristics of a problem.
- **Architectures**: reflex, model-based, goal, utility, learning; **multi-agent systems**.

Module 2 – What you will do



Ant colony

Simulate ants foraging for food.



Flocking birds

Simulate flocking with three simple rules.



Messages between agents

Agents negotiate jobs by passing messages.



Network search

Communicating agents search a network.



Third Mainland Bridge, Lagos: route finding is search

Module 3 – Problem solving by search and heuristics

- Formulating problems; search trees; **major structures** in AI programs.
- **Uninformed search**
- **Heuristic search**: greedy and **A***; local search; constraint satisfaction; problem-solving methods.
- **Games** (minimax, alpha–beta) and **combinatorial explosion**.

Module 3 – What you will do



Missionaries & cannibals

Solve the classic puzzle with search.



Route finder

Lagos to Kano with BFS, uniform-cost and A*.



8-puzzle & 8-queens

Heuristics and local search.



Game AI

Minimax and alpha–beta for tic-tac-toe and Ayo.



Aristotle, the father of logic

Module 4 – Knowledge representation and reasoning

- Knowledge, KR challenges and **KR languages**.
- **Propositional and predicate logic**: English → logic, **clause form**, **resolution**.
- **Semantic networks** (relationships, inheritance, types, components) and **frames**.
- **Non-monotonic** and **probabilistic** reasoning: defaults, Bayes, Bayesian networks.

Module 4 – What you will do

Prove by resolution

Run the classic ‘Marcus’ proof in Python.

Chain rules

Forward and backward chaining on a family tree.

Semantic nets & frames

Inheritance with exceptions; frames with demons.

Bayes in action

What a positive malaria test really means.



A robot lawn mower

Module 5 – NLP, expert systems, AI languages, vision and robots

- **NLP**: language understanding, syntactic and semantic structures, seven practical apps.
- **Expert systems**: characteristics, components, types, requirements, technology, development.
- **Programming languages for AI**: Python.
- **Image recognition** (digits, faces), **simple robots**, **AI animation** and the **lawn-robot** project.

Module 5 – What you will do



NLP group practicals

Spam, sentiment, spelling, QA, translation, speech, chatbot.



Face recognition

Recognise faces with eigenfaces.



Expert system

Build a generator-fault adviser that explains itself.



Robots & animation

Braitenberg vehicles, animations, the lawn robot.

Tools you will use (all free)



Google Colab

Run Python in your browser — nothing to install.



Python

NumPy, scikit-learn, NLTK and more.



NetLogo Web

Agent simulations in the browser.



Teachable Machine

Train image models in the browser.

Books and free resources

Resource	What you'll find	Link
Russell & Norvig, 'AI: A Modern Approach'	The main textbook	aima.cs.berkeley.edu
Rich & Knight, 'Artificial Intelligence'	Classic text: KR, logic, expert systems	Library
Elements of AI	Free online introduction to AI	elementsofai.com
NLTK book	NLP with Python, free	nltk.org/book
Google Colab	Run the course programs	colab.research.google.com

Skills you will leave with



Design intelligent agents

Specify, classify and build agents for real problems.



Choose the right search method

And explain its trade-offs.



Represent and reason with knowledge

Logic, rules, networks, frames, probabilities.



Build practical AI apps

Spam filters, chatbots, face recognition, robots.



Think critically about AI

Its limits, risks and responsibilities.

Where these skills can take you

Machine-learning engineer

Build models that learn from data.

Data scientist

Find patterns and insights in data.

NLP engineer

Language tools, chatbots, translation.

Robotics engineer

Robots for farms, factories, homes.

AI researcher

Push the frontier — including for African languages.

AI entrepreneur

Solve Nigerian problems with AI.